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AVTools System Designer (Standard and Pro)

Version 2.0 – Quick Start & Notes

Thank you for downloading AVTools System Designer by J. Stevens BIM Acoustics. This Revit add-in accelerates distributed ceiling loudspeaker layout while keeping engineering intent clear, deterministic, and transparent. Version 2.0 adds full circuiting, amplifier-to-speaker cabling, and a 2,500+ record acoustic materials database.

1. Installation

1. Unzip the downloaded file.

2. Run the installer (.msi).

3. If Windows shows a SmartScreen warning:

- Click More info → Run anyway

4. Launch Autodesk Revit. Version 2.0 supports Revit 2025 and 2026. For Revit 2022–2024, install AVTools v1.2.1 instead.

2. Where to Find the Tool in Revit

Once Revit is open:

1. Click the BIM Acoustics tab in the top ribbon.

2. Locate the AV Tools panel.

3. Click AV Tools System Design to launch.

Recommended

Start with the project open to a Floor Plan or Reflected Ceiling Plan (RCP) view for the level you want to work on. Several features (Room Acoustics refresh, Draw Wiring) are scoped to the active plan view's level and will show an empty list if you're in a 3D view or schedule.

3. First Launch

On first launch System Designer will:

• Open a Registration dialog asking for your name, company, and email. Your 14-day Standard trial starts when you complete registration. If you have already subscribed, enter the email address you used to subscribe to validate your add-in. You can subscribe later from the About tab — Pro and Standard-tier features stay locked until you do.

• Import a small set of JSBA project parameters into the current project.

• Auto-load the shipped JSBA tag and coordination families so tag dropdowns are populated immediately.

• Create a JSBA Control Data schedule to track placed loudspeakers.

⚠ Do not delete or manually edit the JSBA-generated schedules. They are the internal record the add-in uses to track placed devices.

Using the Schedules for Documentation (Recommended)

• Duplicate any JSBA-generated schedule.

• Modify the duplicate freely: add/remove columns, change formatting, place on sheets, export to reports.

• Leave the original JSBA schedule unchanged so the add-in can continue managing updates.

4. Tab Workflow (Pro)

System Designer is organized around an end-to-end design flow:

Rooms → Configuration → Direct-Field Coverage → Room Acoustics → Circuiting → Amps & Cabling → Coordination → About

You can work host-model or linked-model. Linked-room projects are fully supported for placement, circuiting, and coordination.

Placement Views vs Mapping Views

• Loudspeaker Placement — speakers are placed ceiling-hosted in the model; typically reviewed and coordinated in an RCP.

• Coverage Mapping — ISO-line coverage maps are drawn in the Floor Plan view for the level. New in v2.0: maps are drawn as view-specific detail curves that remain visible in Hidden Line display mode and behind linked models.

How Placement Updates Work (By Design)

Each time you re-run placement with updated inputs:

• Loudspeakers previously placed by the tool are removed

• New loudspeakers are placed from the current calculations

Manual coordination is expected after placement — center speakers in corridors, coordinate with lights/sprinklers/diffusers, and remove individual speakers where coverage is not required.

If you need more speakers than the tool placed, copy/paste an existing tool-placed loudspeaker in the same room. This preserves internal tracking so future recalculation still sees the added devices.

5. Room Acoustics (Standard)

Room Acoustics provides:

• Automated room surface extraction (walls, floors, ceilings, doors, windows, curtain walls) with per-face classification

• Acoustic material assignment via the built-in 2,500+ record material database (bundled) or a cloud-synced language-specific database

• RT60 calculation in three formulas — Sabine, Norris-Eyring, and Arau-Puchades — per octave band from 63 Hz to 8 kHz

• STI estimation from RT60 bands, background noise, speaker directivity, and N-factor

• Critical distance and D/R ratio reporting

• A multi-category Room Acoustics Schedule ready for sheets

Room Acoustics is level-scoped — open a floor plan view at your selected rooms' level, then click Refresh. A clear message appears if the active view doesn't carry a level.

6. Circuiting (Pro)

• Assign circuit IDs per room and zone

• Auto-select the nearest speaker tap at or above the required power

• Write circuit parameters (ID, quantity, impedance, circuit watts) to each speaker

• Create a JSBA Loudspeaker Circuit Schedule filtered to JSBA-managed devices

• Draw Wiring — lays down cable paths and homerun notes in the floor plan

• Place ID Tags — multi-category tags at a selectable placement offset (Top-Right, Center, etc.)

7. Amps & Cabling (Pro)

• Place equipment racks in your project where they will be located (AV Closets, IDF's, etc.). Discover amp racks placed in the model and assign them automatically via nearest-neighbor

• Report line loss, damping factor, and cable length estimates per circuit, with warning thresholds

· Configurable wire-gauge defaults (16 AWG → 10 AWG)

• Assign Rack IDs — writes room number/name back to each rack

• Place Rack Tags — multi-category tags for racks

8. Coordination (Pro)

• Clash detection between loudspeakers and MEP elements (lighting, diffusers, sprinklers) with four clearance presets: Physical, Tight, Normal, Conservative

• Per-category clearance tolerances with AABB intersection tests

• Issue tracking with Open / Coordination / Resolved statuses

• Places JSBA Coordination Issue Marker families at conflict locations so architects and MEP can see the flagged items in the model

• Tab optionally locks downstream work until unresolved conflicts are cleared

9. Acoustic Materials Database

• 2,500+ records shipped bundled with the add-in (works offline, no login required)

• Optional cloud sync per language — new languages available as the database grows

• Assign Materials dialog filters by room surface, matches Revit surface types to acoustic materials, and applies α-coefficients across all eight octave bands

• Custom user materials are supported via the User Materials panel

10. Lucius AI Assistant

Lucius is integrated directly as a dockable pane:

• Chat + voice interaction

• Navigate the workflow step-by-step

• Help understand spacing models, coverage, and design intent

• Pro only: live model-editing via tool use — Lucius can assign materials, help triage coordination issues, and run audits on the current project

Transcript Opt-In (Optional)

You may opt in to anonymous transcript capture to help improve Lucius. No project geometry or identifying information is collected. Configurable from the About tab at any time.

11. Trial & Licensing

• 14-day Standard trial starts on first registration

• Two subscription tiers available from the About tab:

• Standard — $60/mo or $600/yr — layout, coverage, Room Acoustics, materials, Lucius chat/voice

• Pro — $99/mo or $990/yr — everything in Standard plus circuiting, amps & cabling, ID/rack tags, homerun notes, and Lucius live model edits

• After subscribing, enter your email on the About tab and click Validate Now

12. What's New in v2.0

· Full Pro workflow (circuiting → amps & cabling → coordination)

• Multi-category Room Acoustics schedule with lead-speaker carrier

• 2,500+ record bundled acoustic materials database with cloud sync per language

• Registration + 14-day Standard trial

• Iso-coverage maps now draw as detail curves — visible in Hidden Line mode and behind linked models

• Room Acoustics supplement scan for doors, windows, and curtain walls clipped to room bounding boxes

• Startup load of shipped JSBA tag + coordination families

• Revit 2025.3+ tile centering; Revit 2026 API compatibility

• Linked-room support across placement, circuiting, and coordination

13. Support & Feedback

Support email: support@bimacoustics.net

When reporting an issue, please include:

• Revit version (2025 or 2026)

• Tier (Standard / Pro / Trial)

• Active view type (RCP vs Floor Plan, 3D, etc.)

• Steps performed

· Log bundle (available from the About tab → Bundle Logs)

Thank you for being an early user. Your feedback directly shapes future versions.

— Jerrold Stevens

J. Stevens BIM Acoustics

https://www.bimacoustics.net